Resume Screening & Candidate Shortlisting

Applied ML & Intelligent Decision Systems Suite (2025) · Notebook 4 of 4

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0.7839	0.5449	19,158	TF-IDF
ROC-AUC	**F1 Score**	**Records**	**Features**
Best model score	Seeking class F1	Candidate profiles	Text + structured

1. Project Overview

Recruitment is one of the most resource-intensive processes in any organisation. Manually screening hundreds of resumes to identify candidates who are actively seeking a role change is time-consuming and subject to human bias. This notebook builds a machine learning pipeline to predict whether a data science candidate is actively looking for a job change, enabling HR teams to prioritise outreach and reduce time-to-hire.

The dataset used is the Kaggle HR Analytics: Job Change of Data Scientists dataset, comprising 19,158 candidate profiles collected from a data science training platform. Each record contains 13 features spanning candidate demographics, education background, work experience, and company characteristics. The target variable indicates whether the candidate is actively seeking a new position (1) or not (0). What distinguishes this notebook from the rest of the suite is its mixed feature space: structured numeric and categorical fields combined with a text feature (major_discipline), processed using TF-IDF vectorisation and merged via sparse matrix stacking before modelling.

2. Methodology

2.1 Data Preprocessing

The dataset presented moderate missing values across several columns, requiring careful imputation before feature engineering:

- The enrollee_id column was dropped as a unique identifier with no predictive value.
- The major_discipline text column had missing values filled with the string 'unknown', allowing TF-IDF to treat it as a valid token rather than raising a null error.
- Remaining categorical columns (gender, enrolled_university, education_level, company_size, company_type, experience, last_new_job) were imputed using the mode (most frequent value) of each column.
- All categorical structured features were label-encoded to integers, then standardised using StandardScaler to normalise the feature ranges before combining with TF-IDF output.

2.2 TF-IDF Feature Engineering

The major_discipline column contains free-text discipline labels describing a candidate's academic background (e.g., 'STEM', 'Business Degree', 'Humanities'). Rather than treating this as a single categorical variable via label encoding, TF-IDF (Term Frequency-Inverse Document Frequency) was applied to extract a richer numeric representation:

- TfidfVectorizer with max_features=50 was fitted on the major_discipline column, producing a 50-dimensional sparse vector per candidate.

- The TF-IDF sparse matrix was horizontally stacked with the scaled structured features using scipy.sparse.hstack, producing a unified feature matrix of shape (19,158, structured_cols + 50).
- This approach mirrors real-world resume parsing pipelines where unstructured text fields (skills, descriptions, disciplines) are vectorised and merged with structured profile attributes.
- The fitted TF-IDF vectorizer was saved as a .pkl artifact alongside the model and scaler, since all three are required in sequence for any future inference.

2.3 Models Trained

Model	ROC-AUC	F1 Score	Notes
Logistic Regression	~0.74	~0.49	Linear baseline
Random Forest	~0.77	~0.53	Strong second
XGBoost	**0.7839**	**0.5449**	**Best model — selected**

XGBoost achieved the best performance across both metrics and was selected as the production model. Three artifacts were saved for deployment: the trained XGBoost model, the StandardScaler, and the TfidfVectorizer — all three must be applied in the same order for any future inference to be valid.

3. Key Findings

3.1 Contextualising the Performance Scores

A ROC-AUC of 0.7839 and F1 of 0.5449 are consistent with published benchmarks on this specific dataset, which is widely recognised in the ML community as a challenging prediction problem. The core difficulty is not data quality but signal availability: whether a professional is privately considering a career move is a psychographic state that is poorly approximated by observable structural features like company size, education level, or training hours. The model captures statistically significant patterns but cannot observe latent intent — a fundamental limitation of any passive data-driven recruitment system.

3.2 Key Predictors from EDA and SHAP

Exploratory analysis and SHAP explainability both converged on the same top predictors. City development index was consistently the strongest feature: candidates from cities with lower development indices — indicating smaller job markets and fewer local opportunities — are significantly more likely to seek a role change. This aligns with economic theory around labour mobility. Training hours completed on the platform showed moderate importance, with candidates who completed more training being more likely to be actively seeking a new role, suggesting self-improvement signals career transition intent. Company size and company type showed clear patterns: candidates at very large companies and startups had differing job-seeking rates, reflecting the different career dynamics of those environments. Among the TF-IDF features derived from major_discipline, STEM was the highest-importance token, reflecting its prevalence and relevance in data science hiring.

3.3 TF-IDF as a Pipeline Design Choice

Using TF-IDF on the major_discipline column — rather than simply label-encoding it as a categorical — demonstrates a key NLP feature engineering pattern applicable to real resume screening systems. In practice, resume fields such as skills, job titles, and degree descriptions contain rich text that label encoding cannot capture. The TF-IDF approach treats each unique term as a feature dimension and weights it by its discriminative power across the corpus. Combined with scipy.sparse.hstack for memory-efficient matrix merging, this pipeline mirrors the architecture of production-grade candidate screening systems. The vectorizer is saved as a separate artifact precisely because it encodes vocabulary from the training data and must be reused — not re-fitted — at inference time.

4. Technical Stack

Category	Tools / Libraries
Language	Python 3.10+
ML Framework	Scikit-learn, XGBoost
Text Vectorisation	TF-IDF (TfidfVectorizer, max_features=50)
Matrix Operations	scipy.sparse (hstack, csr_matrix)
Imbalance Handling	imbalanced-learn (SMOTE)
Feature Scaling	StandardScaler on structured features
Explainability	SHAP (TreeExplainer, max_display=15)
Evaluation	ROC-AUC, F1-score, Classification Report, Confusion Matrix
Visualisation	Matplotlib, Seaborn

Saved Artifacts	recruitment_model.pkl, recruitment_scaler.pkl, recruitment_tfidf.pkl
Dataset	Kaggle HR Analytics: Job Change of Data Scientists (19,158 records, 13 features)

Suite Summary

Notebook	Domain	Best Model	ROC-AUC	Key Metric
NB 1	Healthcare	Random Forest	0.6613	ROC-AUC
NB 2	Fraud Detection	Random Forest	0.9740	PR-AUC 0.8686
NB 3	Education	XGBoost	0.9279	F1 0.8167
NB 4	Recruitment	XGBoost	0.7839	F1 0.5449